

## 02 S3 to SNS Automated Notification

### Lab 2: S3 to SNS Automated Notification

This lab demonstrates how to configure an S3 bucket to publish automatic notifications to an SNS topic upon object creation, which then delivers the message to an email subscriber.

#### 1. Objective

Send an automated email notification via AWS SNS whenever a new object is created in an S3 bucket.

#### 2. Prerequisites

- An AWS account with console access.
- An email address to receive notifications.
- Estimated Time: 15 minutes

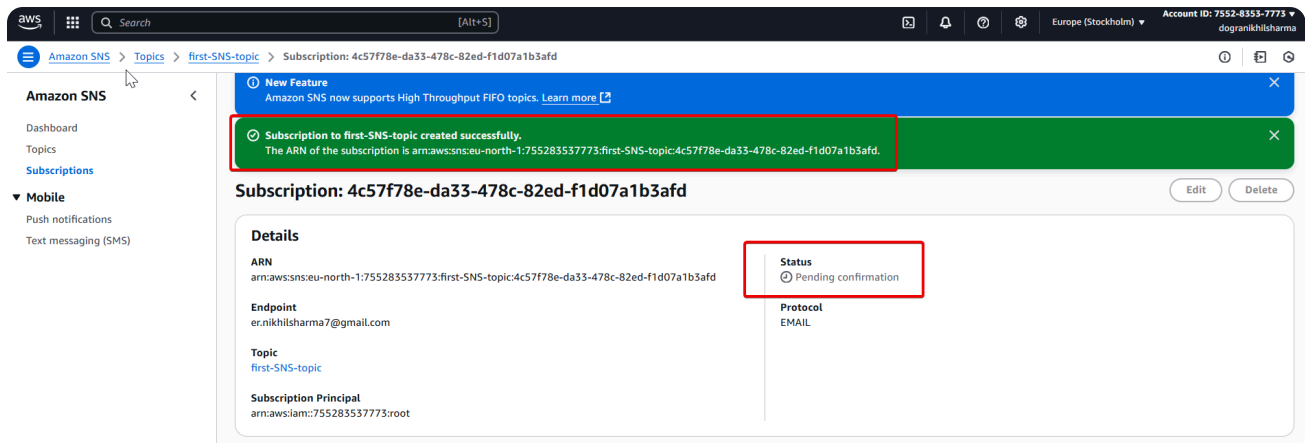
#### 3. Detailed Steps

##### Step 1: Create an SNS Topic

1. Navigate to **SNS** in the AWS Console.
2. Click **Topics** -> **Create topic**.
3. Choose **Standard** for the topic type.
4. Name the topic: `first-SNS-topic` (use hyphens instead of underscores for better compatibility).
5. **Access policy**: Leave as **Only the topic owner can publish messages** for now. We'll configure S3 permissions separately for better security practice.
6. Leave other settings as default and click **Create topic**.
7. **Important**: Copy the **Topic ARN** from the topic details page - you'll need this later.

##### Step 2: Create an Email Subscription

1. In your `first-SNS-topic` overview, click **Create subscription**.
2. **Protocol**: Select **Email**.
3. **Endpoint**: Enter your email address (e.g., `your_email@example.com`).
4. Leave **Subscription filter policy** empty for now.
5. Click **Create subscription**.
6. **Confirm Subscription**:
  - You will receive an email from AWS Notifications with the subject "AWS Notification - Subscription Confirmation".
  - Open the email and click the **Confirm subscription** link.
  - Verify the status changes to **Confirmed** in the SNS console.



### Subscription: 4c57f78e-da33-478c-82ed-f1d07a1b3afd

#### Details

##### ARN

arn:aws:sns:eu-north-1:755283537773:first-SNS-topic:4c57f78e-da33-478c-82ed-f1d07a1b3afd

##### Endpoint

er.nikhilsharma7@gmail.com

##### Topic

[first-SNS-topic](#)

##### Subscription Principal

arn:aws:iam::755283537773:root

##### Status

Confirmed

##### Protocol

EMAIL

## Step 3: Create a New S3 Bucket

1. Navigate to **S3** in the AWS Console.
2. Click **Create bucket**.
3. Name the bucket: `bucket-for-sns-lab-[YOUR-INITIALS]` (ensure it's globally unique by adding your initials or a random number).
4. Choose your desired AWS Region (same region as your SNS topic for optimal performance).
5. Leave other settings as default and click **Create bucket**.

## Step 4: Configure SNS Topic Policy for S3 Access

**Security Configuration** This step is crucial – S3 needs explicit permission to publish to your SNS topic.

1. Go back to your SNS topic ( `first-SNS-topic` ).
2. Click on the **Access policy** tab.
3. Click **Edit** to modify the policy.
4. Replace the existing policy with the following JSON (replace `YOUR-TOPIC-ARN` and `YOUR-BUCKET-NAME` with actual values):

```
{
  "Version": "2012-10-17",
  "Id": "S3-SNS-Policy",
  "Statement": [
    {
      "Sid": "AllowAccountOwner",
```

```

    "Effect": "Allow",
    "Principal": {
      "AWS": "arn:aws:iam::755283537773:root"
    },
    "Action": [
      "SNS:Publish",
      "SNS:RemovePermission",
      "SNS:SetTopicAttributes",
      "SNS>DeleteTopic",
      "SNS:ListSubscriptionsByTopic",
      "SNS:GetTopicAttributes",
      "SNS:AddPermission",
      "SNS:Subscribe"
    ],
    "Resource": "arn:aws:sns:eu-north-1:755283537773:first-SNS-topic"
  },
  {
    "Sid": "Allows3Publish",
    "Effect": "Allow",
    "Principal": {
      "Service": "s3.amazonaws.com"
    },
    "Action": "SNS:Publish",
    "Resource": "arn:aws:sns:eu-north-1:755283537773:first-SNS-topic",
    "Condition": {
      "StringEquals": {
        "aws:SourceAccount": "755283537773"
      },
      "ArnEquals": {
        "aws:SourceArn": "arn:aws:s3:::YOUR-BUCKET-NAME"
      }
    }
  }
]
}

```

5. **Important:** Replace 123456789012 with your actual AWS Account ID (found in the top-right corner of AWS Console).
6. Click **Save changes**.

## Step 5: Configure S3 Event Notification to SNS

1. Navigate to your S3 bucket ( bucket-for-sns-lab-[YOUR-INITIALS] ).
2. Go to the **Properties** tab.
3. Scroll down to **Event notifications** and click **Create event notification**.
4. **Event name:** SNS-Notification .
5. **Prefix:** Leave empty (optional: add a prefix if you want to filter specific folders).
6. **Suffix:** Leave empty (optional: add .csv if you only want CSV file notifications).
7. **Event types:** Select **All object create events** or specifically **s3:ObjectCreated:Put**.
8. **Destination:** Select **SNS topic**.

9. Choose **Choose from your SNS topics** and select your `first-SNS-topic`.
10. Click **Save changes**.

**Potential Error** If you get a permission error, ensure you completed Step 4 (SNS topic policy configuration) correctly with the right ARNs and Account ID.

## 4. Verification (Lab 2)

### Step 1: Upload a File to S3

1. Navigate to your S3 bucket ( `bucket-for-sns-lab-[YOUR-INITIALS]` ).
2. Click **Upload**, then **Add files**.
3. Select any file (e.g., `storedata.csv` ) and click **Upload**.

### Step 2: Check Email Notification

1. Open the email inbox for the subscribed email address.
2. You should receive an email within 1-2 minutes with the subject "Amazon S3 Notification".
3. The message body will contain a detailed JSON notification of the S3 object creation event, including:
  - Bucket name
  - Object key (filename)
  - Event time
  - Object size
  - AWS region

### Step 3: Verify SNS Metrics (Optional)

1. Go to your SNS topic in the console.
2. Check the **Monitoring** tab to see message publish metrics.
3. Verify that messages were successfully published and delivered.

## 5. Testing Variations

### Test Different File Types

- Upload different file types (.txt, .jpg, .pdf) to see if all trigger notifications.
- Try uploading to different folders within the bucket.

### Test Multiple Subscribers

- Add an SMS subscription to the same topic.
- Upload a file and verify both email and SMS notifications are received.

## 6. Cleanup (Lab 2)

### Step 1: Delete S3 Bucket Event Notification

1. In your S3 bucket, go to **Properties** -> **Event notifications**.
2. Select `SNS-Notification` and click **Delete**.

## Step 2: Delete SNS Subscription

1. In the SNS console, go to **Subscriptions**.
2. Select your email subscription and click **Delete**.
3. Confirm the deletion.

## Step 3: Delete SNS Topic

1. In the SNS console, go to **Topics**.
2. Select `first-SNS-topic` and click **Delete**.
3. Type "delete me" to confirm deletion.

## Step 4: Delete S3 Bucket

1. **First, empty the bucket:** Select all objects and click **Delete**.
2. In the S3 console, select your bucket and click **Delete**.
3. Type the bucket name to confirm deletion.

## 7. Troubleshooting

### Common Issues:

1. **No Email Received:**
  - Check spam/junk folder
  - Verify email subscription is confirmed
  - Check SNS topic policy allows S3 to publish
2. **Permission Errors:**
  - Ensure SNS topic policy includes correct S3 service principal
  - Verify AWS Account ID is correct in the policy
  - Check bucket ARN matches exactly
3. **S3 Event Notification Setup Fails:**
  - Ensure SNS topic and S3 bucket are in the same region
  - Verify the topic ARN is correct
  - Check that the SNS topic exists and is accessible

### Verification Steps:

1. Test the setup by uploading a small text file first.
2. Check CloudWatch logs for any error messages.
3. Verify all ARNs and Account IDs are correct in policies.